KIT - Kalaignarkarunanidhi Institute of Technology

(An Autonomous Institute)

Department of Computer Applications

M23CAP401– PROJECT

REVIEW 1

GUIDED BY,
Dr E .VIJAYAKUMAR MCA.,Ph.D.,
ASSOCIATE PROFESSOR,
HEAD OF DEPARTMENT.

PRESENTED BY,
GANESH.A(711524MMC010),
DEPARTMENT OF MCA.

PROJECT TITLE:

Smart Answering System using Retrieval-Augmented Generation (RAG)

ABSTRACT

- Developed a Smart Answering System using Retrieval-Augmented Generation (RAG).
- Understands user questions in natural language.
- Retrieves relevant information from uploaded documents before generating answers.
- Converts documents into vectors for efficient semantic search.
- Stores vectors in a vector database for fast information retrieval.
- Uses an open-source Large Language Model (LLM) to generate context-based responses.
- Reduces hallucination by grounding answers in retrieved documents.
- Handles large volumes of data efficiently.
- Suitable for chatbots, document search systems, and knowledge management applications.

EXISTING SYSTEM

- Uses only keyword-based search without semantic understanding.
- Primarily works with plain text and struggles to process PDFs or images effectively.
- Stores data in traditional databases without contextual understanding.
- Generates answers without validating information, leading to possible incorrect responses.
- Cannot efficiently handle large unstructured data such as scanned documents.
- Performance decreases as data size increases.
- Requires manual rule-based methods for information extraction.
- Provides less accurate answers for complex or multi-part queries.
- Not suitable for real-time intelligent applications.
- Does not utilize modern AI techniques or multimodal models.

PROPOSED SYSTEM

- Can understand text, images, PDFs, and scanned documents in one system
- Supports different file formats and adjusts automatically
- Uses open-source AI models with a smart agent to understand user questions
- Converts all data into vector form so the system can understand meaning
- Stores the data in a vector database for quick access
- Smart agent analyzes and improves the user question if needed
- Finds the most relevant information based on the question
- Thinks step-by-step to generate accurate answers
- Gives clear, explainable, and context-based responses
- Useful for chatbots, research, document analysis, and decision making

SYSTEM MODULES

- Modules 1- Document Ingestion & Management Module
- Modules 2- Multimodal Processing & Embedding Module
- Modules 3- Vector Storage & Retrieval Module
- Modules 4- Agentic Query Understanding Module
- Modules 5 - Context Reasoning & Answer Generation Module

Module description

Modules 1-document ingestion & management

- Uploads pdfs, images, and text files
- Extracts and organizes document content
- Maintains document metadata for structured processing

Modules 2-multimodal processing & embedding

- Processes text and image-based content and converts it into numerical vectors (embeddings).
- Enables semantic understanding for better comparison and similarity search.

Modules 3 -vector storage & retrieval

- Stores generated embeddings in a vector database for efficient searching.
- Retrieves the most relevant information based on the semantic similarity of user queries

Modules 4-Agentic Query Understanding

- Analyzes user intent and rewrites complex or unclear queries if necessary.
- Plans intelligent retrieval steps to improve answer accuracy.

Moduels 5-Context Reasoning & Answer Generation

- Combines retrieved information with the user query to generate structured and context based responses.
- Ensures answers are meaningful, accurate, and grounded in the uploaded documents.

MODELS

- **Large Language Model (LLM):**

  Open AI

- **Embedding Model:**

  Sentence Transformers (all-MiniLM-L6-v2)

- **Vision Model:**

  CLIP (Contrastive Language–Image Pretraining)

- **OCR Model:**

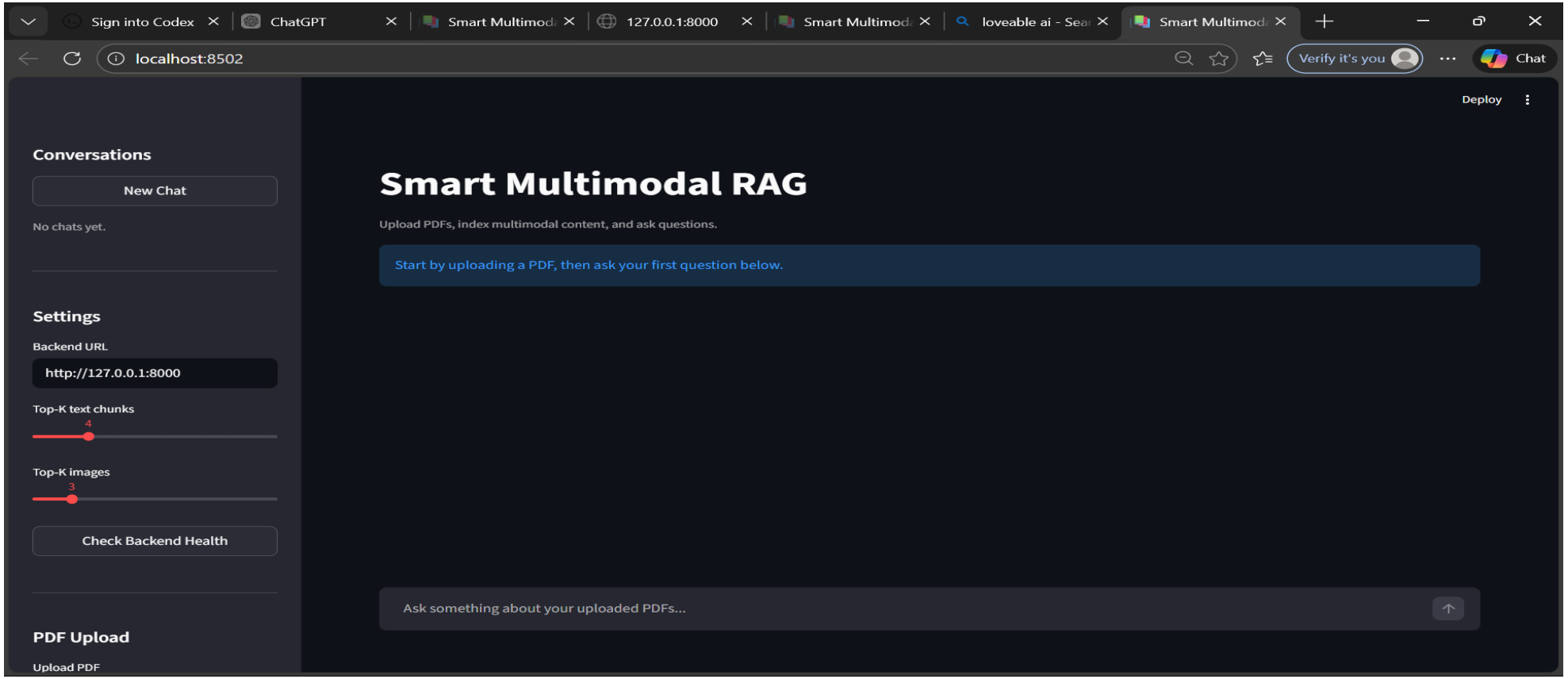
  Tesseract OCR / EasyOCR

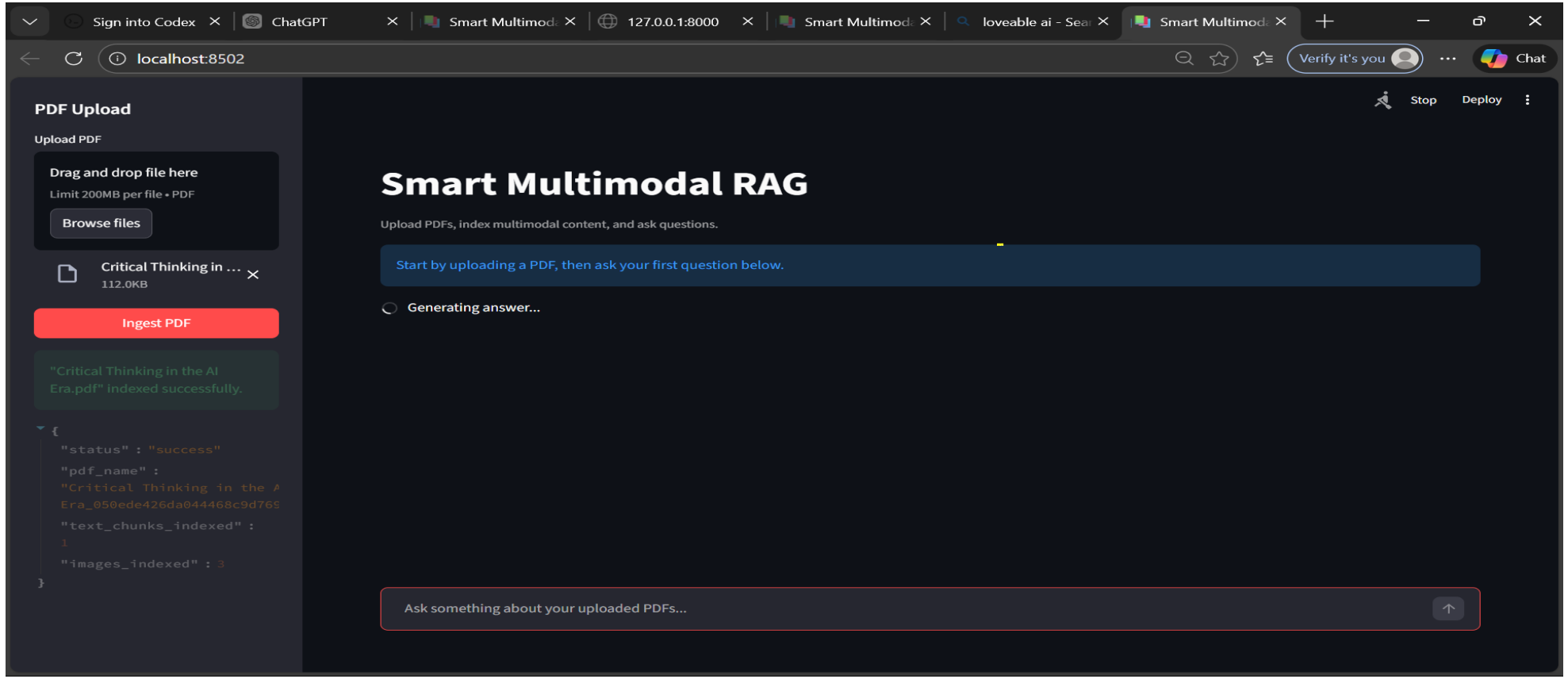
SYSTEM SPECIFICATION

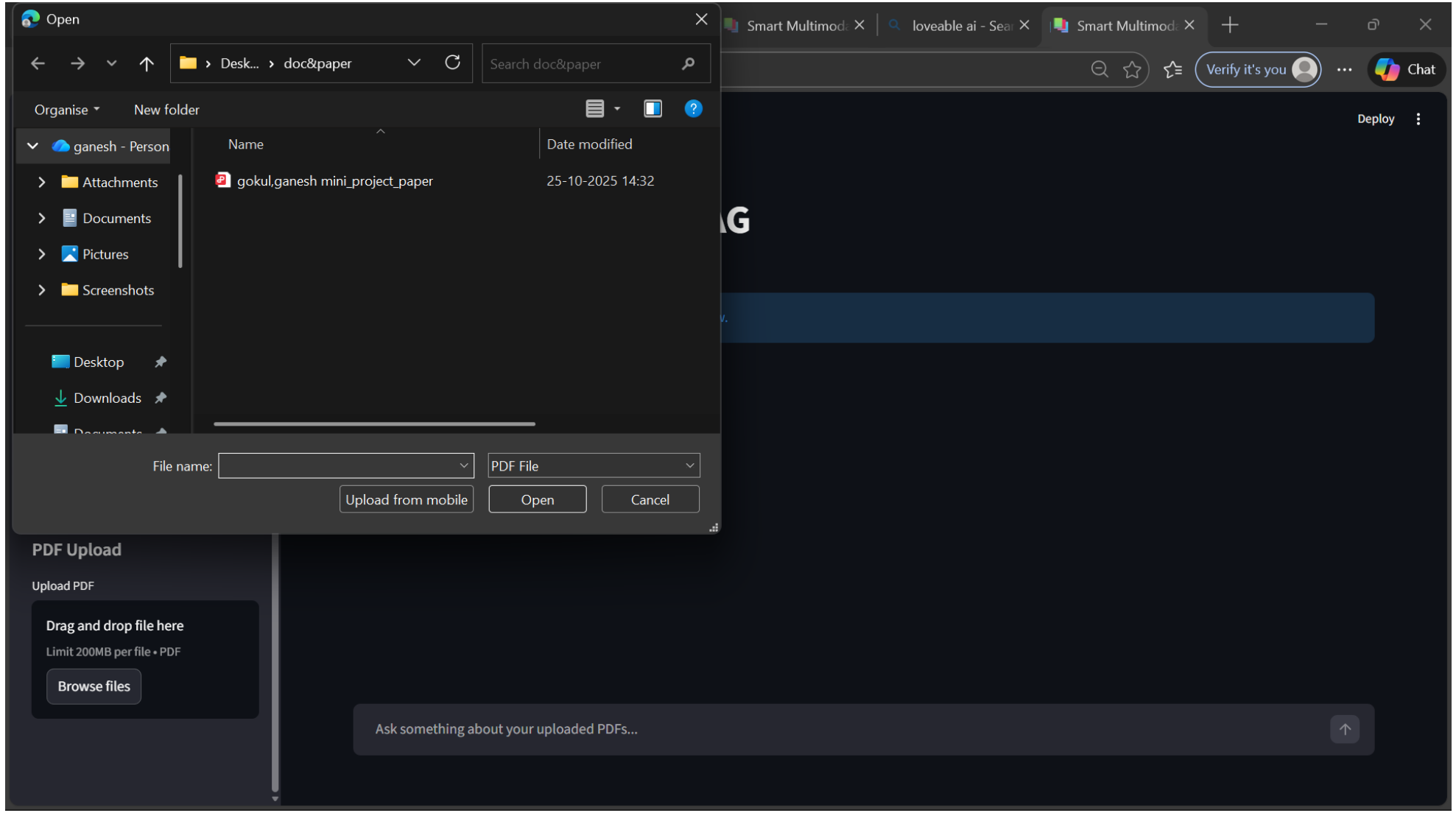
- **HARDWARE**
- Processor: Intel Core i5 or higher
- RAM: Minimum 8 GB (16 GB recommended)
- Storage: Minimum 256 GB HDD/SSD
- System Type: 64-bit computer

- **SOFTWARE**
- Operating System: Windows 11, Linux, or macOS
- Programming Language: Python 3.10 or above
- Database: Chroma Vector Database
- Tools: VS Code

screenshot


Sign into Codex
ChatGPT
Smart Multimod
127.0.0.1:8000
Smart Multimod
loveable ai - Sea
Smart Multimod
localhost:8502
Verify it's you
Chat
Deploy
Conversations
New Chat
No chats yet.
Settings
Backend URL
http://127.0.0.1:8000
Top-K text chunks
4
Top-K images
3
Check Backend Health
PDF Upload
Upload PDF
Smart Multimodal RAG
Upload PDFs, index multimodal content, and ask questions.
Start by uploading a PDF, then ask your first question below.
Ask something about your uploaded PDFs...


Sign into Codex
ChatGPT
Smart Multimod
127.0.0.1:8000
Smart Multimod
loveable ai - Sea
Smart Multimod
localhost:8502
Verify it's you
Chat
Stop
Deploy
PDF Upload
Upload PDF
Drag and drop file here
Limit 200MB per file • PDF
Browse files
Critical Thinking in ...
112.0KB
Ingest PDF
"Critical Thinking in the AI Era.pdf" indexed successfully.
{
"status" : "success"
"pdf_name" :
"Critical Thinking in the A
Era_050ede426da044468c9d769
"text_chunks_indexed" :
1
"images_indexed" : 3
}
Smart Multimodal RAG
Upload PDFs, index multimodal content, and ask questions.
Start by uploading a PDF, then ask your first question below.
Generating answer...
Ask something about your uploaded PDFs...


Open
Desk... › doc&paper
Search doc&paper
Organise
New folder
ganesh - Person
Attachments
Documents
Pictures
Screenshots
Desktop
Downloads
Name
Date modified
gokul,ganesh mini_project_paper
25-10-2025 14:32
File name:
PDF File
Upload from mobile
Open
Cancel
Smart Multimod
loveable ai - Sea
Smart Multimod
Verify it's you
Chat
Deploy
PDF Upload
Upload PDF
Drag and drop file here
Limit 200MB per file • PDF
Browse files
Ask something about your uploaded PDFs...

THANK YOU!